# BrewLite | Deployment Diagram

ASSIGNMENT 2 - TARGET DESIGN | Vercel Hobby + Render + VNPay Sandbox | Live verification pending

PUBLIC INTERNET / HTTPS

3 Browser -> signed VNPay payment URL (HTTPS)

**<<device>>**
**Customer browser**

UI from Vercel
JWT on protected API calls
No backend signing keys

1 / 4 HTTPS :443
UI + REST/JSON

**<<executionEnvironment>> Vercel Hobby**
**Public frontend HTTPS :443**

<<artifact>> Next.js frontend
Static assets + server Route Handler
/api/backend/* proxy + /payment

Vercel environment
NEXT_PUBLIC_API_URL (public)
BACKEND_INTERNAL_URL (server only)
Both target public Render HTTPS

4 Return -> browser
Then HTTPS GET /payment

**<<external service>>**
**VNPay Sandbox**

Hosted payment page
Server-to-server IPN
QueryDR API
Merchant settings: Return / IPN

6 QueryDR
Outbound HTTPS POST
Verified gateway result

5 IPN: server-to-server
HTTPS GET
/payments/vnpay/ipn

2 REST / JSON over public HTTPS :443
Vercel proxy / SSR -> Render ingress

CD
HTTPS

**<<external node>>**
**GitHub / Actions runner**
**PLANNED - no workflow yet**

Repository -> CI checks
Build + test on isolated CI DB
CD only after checks pass

GitHub Actions secrets (planned)
VERCEL_TOKEN + project IDs
RENDER_DEPLOY_HOOK_URL
No live DB/payment secrets

CD / HTTPS

Public HTTPS :443 / TLS ingress -> process $PORT
Render health: GET / -> 200 (liveness only)

**<<executionEnvironment>> Render Web Service**

<<artifact>> NestJS + Prisma Client (dist/main.js)
IPN + shared finalization
Order expiration + reconciliation timers

Render server environment - secrets
DATABASE_URL, JWT_SECRET, INTERNAL_SECRET
VNPAY_HASH_SECRET (values never shown)

PostgreSQL / TCP
:5432, private
Internal DB URL

**Render workspace / region: TBD**

**RENDER PRIVATE NETWORK**
**PostgreSQL is not a public API**

<<executionEnvironment>>
Render PostgreSQL Free

<<artifact>> schema + persisted data
Internal Database URL only
External access OFF (verify)
Same workspace / region as API

## READ THIS AS A TARGET DESIGN

Solid arrows: target runtime communication (verify after deploy).
Purple dashed arrows: planned CI/CD deployment dependencies.
Green boundary: private DB path; no browser / Vercel DB link.
Secrets: Render server env; CI deployment keys in GitHub Secrets.
Return is navigation only. IPN / QueryDR verify before finalization.
Crossing lines without a junction do not indicate a connection.

## OPEN ACCEPTANCE GATES

Live URLs / region / tier / environment: TBD.
Payment status route missing; full Return UI flow blocked.
Free Web tier (if chosen): sleep affects IPN + schedulers.
DB Free lifecycle + private migration runner: verify.
QueryDR Sandbox E2E: pending (see ADR).
PR review / merge into dev: user authorization required.